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`https://github.com/surabhi84ard/Mini/tree/8e9042b13304f4b4923d330711d1eb424065bccb/avr-gcc`

Abstract

This manual shows how to use the 7447 BCD–Seven Segment Display decoder to learn Boolean logic.

1 Components

Component	Value	Quantity
Resistor	220 Ohm	1
Arduino	UNO	1
Seven Segment Display	–	1
Decoder	7447	1
Jumper Wires	M-M	20
Breadboard	–	1

Table 1: Required Components

2 Hardware Setup

2.1 7447 to Display Connections

7447 Output	Display Segment
`b, c, d, e, f, g`	`a, b, c, d, e, f, g`

Table 2: Decoder to Display Mapping

BCD Input	Binary	Decimal Output
0000	0	0
0001	1	1

Table 3: Initial Truth Table

2.2 BCD Input to Decimal Output

3 Software Implementation

3.1 Arduino Pin Mapping

7447 Input	Arduino Pin
D	5
C	4
B	3
A	2

Table 4: Arduino to 7447 Connections

3.2 Boolean Logic

The logic for output D is given by:

$$D = WXYZ + WXYZ'$$

Where: - $WXYZ'$ corresponds to input 0111 - $W'X'Y'Z$ corresponds to input 1000

3.3 Truth Table for Increment Logic

Note

Z	Y	X	W	D	C	B	A	Decimal
0	0	0	0	0	0	0	1	1
0	0	0	1	0	0	1	0	2
0	0	1	0	0	0	1	1	3
0	0	1	1	0	1	0	0	4
0	1	0	0	0	1	0	1	5
0	1	0	1	0	1	1	0	6
0	1	1	0	1	0	0	1	7
0	1	1	1	1	0	1	0	8
1	0	0	0	0	0	0	0	0

Table 5: Incremented Output Truth Table